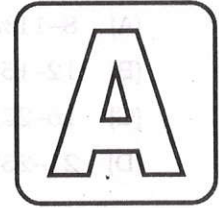


TOU/CE/DAUH/25/10

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO**Test Booklet No. :****02241****TEST BOOKLET
CIVIL ENGINEERING****Series****Time Allowed : 2 Hours****Full Marks : 100****Read the following instructions carefully before you begin to answer the questions :**

1. The name of the Subject, Roll Number as mentioned in the Admission Certificate, Test Booklet No. and Series are to be written legibly and correctly in the space provided on the Answer-Sheet with Black/Blue ballpoint pen.
2. **Answer-Sheet without marking Series as mentioned above in the space provided for in the Answer-Sheet shall not be evaluated.**
3. All questions carry equal marks.

The Answer-Sheet should be submitted to the Invigilator.

Directions for giving the answers : Directions for answering questions have already been issued to the respective candidates in the 'Instructions for marking in the OMR Answer-Sheet' along with the Admit Card and Specimen Copy of the OMR Answer-Sheet.

Example :

Suppose the following question is asked :

The capital of Bangladesh is

- (A) Chennai
(B) London
(C) Dhaka
(D) Dhubri

You will have four alternatives in the Answer-Sheet for your response corresponding to each question of the Test Booklet as below :

(A) (B) (C) (D)

In the above illustration, if your chosen response is alternative (C), i.e., Dhaka, then the same should be marked on the Answer-Sheet by blackening the relevant circle with a Black/Blue ballpoint pen only as below :

(A) (B) (C) (D)

The example shown above is the only correct method of answering.

4. Use of eraser, blade, chemical whitener fluid to rectify any response is prohibited.
5. Please ensure that the Test Booklet has the required number of pages (16) and 100 questions immediately after opening the Booklet. In case of any discrepancy, please report the same to the Invigilator.
6. No candidate shall be admitted to the Examination Hall/Room 20 minutes after the commencement of the examination.
7. No candidate shall leave the Examination Hall/Room without prior permission of the Supervisor/Invigilator. No candidate shall be permitted to hand over his/her Answer-Sheet and leave the Examination Hall/Room before expiry of the full time allotted for each paper.
8. No Mobile Phone, Electronic Communication Device, etc., are allowed to be carried inside the Examination Hall/Room by the candidates. Any Mobile Phone, Electronic Communication Device, etc., found in possession of the candidate inside the Examination Hall/Room, even if on off mode, shall be liable for confiscation.
9. No candidate shall have in his/her possession inside the Examination Hall/Room any book, notebook or loose paper, except his/her Admission Certificate and other connected papers permitted by the Commission.
10. Complete silence must be observed in the Examination Hall/Room. No candidate shall copy from the paper of any other candidate, or permit his/her own paper to be copied, or give, or attempt to give, or obtain, or attempt to obtain irregular assistance of any kind.
11. This Test Booklet can be carried with you after answering the questions in the prescribed Answer-Sheet.
12. Noncompliance with any of the above instructions will render a candidate liable to penalty as may be deemed fit.
13. No rough work is to be done on the OMR Answer-Sheet. You can do the rough work on the space provided in the Test Booklet.

N.B. : There will be negative marking @ 0.25 per 1 (one) mark against each wrong answer.**/40-A****[No. of Questions : 100]****SEAL**

1. For a first class brick, the water absorption, when immersed in cold water for 24 hours, should be
 - (A) 8–11% of its dry weight
 - (B) 12–15% of its dry weight
 - (C) 16–20% of its dry weight
 - (D) 21–25% of its dry weight
2. The process of reducing the moisture content from the timber in order to prevent the timber from possible fermentation is
 - (A) seasoning of timber
 - (B) sorting of timber
 - (C) conservation of timber
 - (D) None of the above
3. In the tensile strength test of stones, the split tensile strength, S is expressed as
 - (A) $S = \frac{W}{\pi dL}$
 - (B) $S = \frac{2W}{\pi dL}$
 - (C) $S = \frac{WL}{\pi d^2}$
 - (D) $S = \frac{2WL}{\pi d^2}$

where

W = Applied load

d = Diameter of the specimen

L = Length of the specimen
4. On the basis of modulus of elasticity in bending, for constructional purpose, timbers are classified as Group B if
 - (A) the modulus of elasticity in bending is above 3 kN/mm² and below 5.6 kN/mm²
 - (B) the modulus of elasticity in bending is above 5.6 kN/mm² and below 9.8 kN/mm²
 - (C) the modulus of elasticity in bending is above 9.8 kN/mm² and below 12.5 kN/mm²
 - (D) the modulus of elasticity in bending is above 12.5 kN/mm² and below 15 kN/mm²
5. Depending on the size, sand is classified as medium sand if the fineness modulus (FM) is
 - (A) 3.20–3.50
 - (B) 2.90–3.20
 - (C) 2.60–2.90
 - (D) 2.20–2.60
6. As per the Indian Standards, to check the dimension of bricks, the number of bricks required is
 - (A) 10
 - (B) 15
 - (C) 20
 - (D) 25
7. The constituent of cement which is responsible for initial setting time of cement is
 - (A) dicalcium silicate
 - (B) tricalcium aluminate
 - (C) tricalcium silicate
 - (D) tetracalcium aluminoferrite
8. The role of superplasticizer in cement concrete is to
 - (A) disperse the particles
 - (B) disperse the particles and remove air bubbles
 - (C) disperse the particles, remove air bubbles and to retard setting
 - (D) retard setting

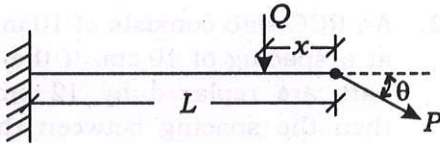
9. The operation of removing humps and hollows of uniform concrete surface is known as

- (A) floating (B) screeding
(C) trowelling (D) finishing

10. Ring and ball apparatus is used for which of the following tests of bitumen?

- (A) Penetration
(B) Viscosity
(C) Softening point
(D) Ductility

11. The maximum bending moment for the beam acted upon by loads as shown in the figure is



- (A) $PL \sin \theta + Q(L - x)$
(B) $PL + Q(L - x)$
(C) $-PL \sin \theta + Q(x - L)$
(D) $-PL \cos \theta - Q(L - x)$

12. A cantilever beam AB of length 4.0 m, fixed at end A, carries a uniformly varying load of intensity zero at B to 2000 N/m at A. The shear force at the middle of the beam, i.e., 2.0 m from ends A and B is

- (A) +4000 N (B) -2000 N
(C) -1000 N (D) +1000 N

13. The deflection due to UDL in a simply supported beam at midpoint is

- (A) $\frac{WL^2}{16EI}$ (B) $\frac{WL^3}{48EI}$
(C) $\frac{WL^3}{24EI}$ (D) $\frac{5WL^4}{384EI}$

14. A beam of length l m carries a uniformly distributed load of w N/m and rests on two simple supports. The cantilever overhang on both sides of the beam is 1.0 m. The bending moment at the centre of the beam is

- (A) $+\frac{wl^2}{2}$ N-m (B) $-\frac{wl^2}{2}$ N-m
(C) $\frac{wl}{2}$ N-m (D) $\frac{wl}{8}(l - 4)$ N-m

15. A beam has a rectangular section of 100 mm $\times$ 200 mm. If it is subjected to a maximum moment of 40 kN-m, the maximum bending stress would be

- (A) 30 N/mm²
(B) 60 N/mm²
(C) 120 N/mm²
(D) 150 N/mm²

16. A cantilever beam of flexural rigidity EI , of length l , with a concentrated load P at the free end, has a strain energy

- (A) $\frac{P^2 l^3}{3EI}$ (B) $\frac{P^2 l^3}{6EI}$
(C) $\frac{Pl^3}{3EI}$ (D) $\frac{P^2 l}{2EI}$

17. For a column of length L , with both ends fixed, the Euler crippling load P is given by

(A) $P = \frac{\pi^2 EI}{4L^2}$

(B) $P = \frac{\pi^2 EI}{L^2}$

(C) $P = \frac{4\pi^2 EI}{L^2}$

(D) $P = \frac{2\pi^2 EI}{L^2}$

The flexural rigidity of the column is EI .

18. A parabolic curved beam of span l and rise h is loaded with a uniformly distributed load of intensity w per unit horizontal length. One end of the beam has a fixed support and the other end is on a horizontal roller support. The equation of the parabolic curve is given by

(A) $y = \frac{2h}{l^2} x(l - x)$

(B) $y = \frac{4l^2}{h} x(l - x)$

(C) $y = \frac{4h}{l} x(l - x)$

(D) $y = \frac{4h}{l^2} x(l - x)$

19. A uniformly distributed load of 16 kN/m covers left half of the span of a three-hinged parabolic arch of span 10.0 m and central rise 2.5 m. The horizontal thrust is

(A) 60 kN (B) 40 kN

(C) 30 kN (D) 20 kN

20. In a suspension bridge, the dominant tensile force is carried by

(A) stiffening girders

(B) suspension cables

(C) hangers

(D) towers

21. A cable is suspended between two points 20 m apart, located at the same level. It carries uniformly distributed load of 16 kN per metre of span. The sag of the cable at mid-span is 4 m. The tension in the cable at the left quarter point is

(A) 200 kN (B) 80 kN

(C) 215.4 kN (D) 297.2 kN

22. An RCC slab consists of 10 mm bars at a spacing of 10 cm. If the 10 mm bars are replaced by 12 mm bars, then the spacing between the bars should be

(A) 10 cm (B) 12 cm

(C) 12.44 cm (D) 14.40 cm

23. A beam AB of span 4 m, fixed at A and B , carries a uniformly distributed load of 1500 N/m. The support A sinks by 1 cm. $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 8000 \text{ cm}^4$. The fixed end moment at end A is

(A) -2000 N-m

(B) +4000 N-m

(C) +6000 N-m

(D) -8000 N-m

24. When a member is subjected to an axial tensile load, the greatest normal stress is equal to
- half the maximum shear stress
 - maximum shear stress
 - twice the maximum shear stress
 - None of the above
25. A reinforced concrete beam 300 mm (width) × 600 mm (depth), subjected to an applied moment of 140 kN-m, is provided with 4 nos. of 25 mm diameter reinforcing bars, at an effective cover of 50 mm. Assuming M20 grade of concrete and Fe415 grade of steel, the stress in steel is
- 131 MPa
 - 141 MPa
 - 151 MPa
 - 161 MPa
26. The neutral axis depth factor of a balanced section depends on the
- permissible tensile stress in steel
 - grade of concrete
 - modular ratio
 - All of the above
27. The maximum spacing for shear reinforcement in a beam is
- greater than 300 mm
 - greater than and equal to 300 mm
 - less than 300 mm
 - less than and equal to 300 mm
28. Where the depth of the web in a beam exceeds 750 mm, the side face reinforcement should not be less than
- 0.1 percent of the web area
 - 0.12 percent of the web area
 - 0.15 percent of the web area
 - 0.85 percent of the web area
29. In limit state of collapse in flexure, the maximum strain in the tension reinforcement in the section at failure shall not be less than
- $0.85 \frac{f_y}{E_s}$
 - $\frac{f_y}{1.15E_s} + 0.002$
 - $\frac{f_y}{1.15E_s} + 0.0035$
 - $0.85 \frac{f_y}{E_s} + 0.015$
30. On either side of the point of stress in a reinforcing bar, the required development length in concrete is
- proportional to the bar diameter
 - proportional to stress level in steel bar at the point
 - inversely proportional to the bond strength of the concrete
 - All of the above

31. The effective cover of beam depends on the

- (A) diameter of main reinforcement
- (B) width of the beam
- (C) depth of the beam
- (D) grade of steel

32. In order that the vertical deflection limit of 10 mm of a simply supported rectangular beam of size 250 mm (width) $\times$ 400 mm (depth) and span 4 m is satisfied, the maximum uniformly distributed load, including its self-weight, is

- (A) 75 kN/m (B) 100 kN/m
- (C) 125 kN/m (D) 150 kN/m

33. For limit state method of design, the permissible bearing stress for transfer of load at the base of column to the top of the supporting footing is

- (A) $0.25f_{ck}$ (B) $0.35f_{ck}$
- (C) $0.45f_{ck}$ (D) $0.55f_{ck}$

34. The thickness at the edge of a pile cap should not be less than

- (A) 150 mm
- (B) 300 mm
- (C) 500 mm
- (D) None of the above

35. While designing a structure under earthquake condition, effects due to vertical earthquake shaking shall be considered when

- (A) the structure is located in seismic zone IV or V
- (B) the structure has vertical or plan irregularities
- (C) the structure is rested on soft soil or the structure has large overhangs of structural members or sub-systems
- (D) any of the above conditions applies

36. By making a moment-resisting frame ductile

- (A) induced seismic force can be reduced
- (B) probability of collapse is reduced
- (C) reversible lateral displacements, after formation of plastic hinges, can be accommodated
- (D) All of the above

37. In a counterfort retaining wall

- (A) the counterforts take reactions only from the heel slab
- (B) the counterforts are subjected to compressive stresses
- (C) the counterforts act as inverted T-beam of varying rib depth
- (D) None of the above

38. As per IS : 456 standard, the average permissible bond stress for plain bars in M25 grade of concrete is
 (A) 0.8 MPa (B) 0.9 MPa
 (C) 1.1 MPa (D) 1.2 MPa
39. According to basic consideration of working stress method for reinforced concrete design, the stress-strain relationship between steel and concrete under working load is
 (A) linear
 (B) parabolic
 (C) logarithmic
 (D) quadratic
40. When Poisson's ratio is 0.2, the relation between E (Young's modulus) and K (bulk modulus) is
 (A) $E = K$ (B) $E = 1.5K$
 (C) $E = 1.8K$ (D) $E = 2K$
41. For a cylindrical water tank, 1.5 m in diameter and 2.0 m in height, if the efficiency of the vertical joints is 95% and the safe maximum hoop stress is 100 N/mm^2 , the thickness required for the tank wall is
 (A) 0.135 mm
 (B) 0.155 mm
 (C) 0.175 mm
 (D) None of the above
42. For a sloped isolated RCC footing, the effective area resisting one-way shear is
 (A) rectangular
 (B) trapezoidal
 (C) polygonal
 (D) cylindrical
43. If the allowable shear stress in weld is 90 N/mm^2 , the strength of 5 mm fillet weld per metre length would be ($k = 0.7$)
 (A) 250 kN (B) 315 kN
 (C) 415 kN (D) 520 kN
44. Los Angeles abrasion test is used for aggregates to check
 (A) strength
 (B) toughness
 (C) hardness
 (D) water absorption
45. The minimum sight distance required to avoid head-on collision of two cars approaching from the opposite directions at 90 kmph and 60 kmph, assuming a reaction time of 2.5 s, coefficient of friction of 0.7 and a brake efficiency of 50 percent in both the cases, is
 (A) 169.98 m (B) 235.8 m
 (C) 153.8 m (D) 82.2 m

46. The radius of a horizontal curve is 100 m. The design speed is 50 kmph and the design coefficient of lateral friction is 0.15. If the full lateral friction is assumed to develop, the required superelevation is

- (A) 0.917 (B) 0.524
(C) 0.234 (D) 0.047

47. The penetration test of bitumen is used to determine

- (A) viscosity
(B) grade
(C) ductility
(D) softening point

48. The psychological widening of a pavement on horizontal curve of radius 230 m for a ruling speed 80 kmph is

- (A) 0.255 m (B) 0.185 m
(C) 0.55 m (D) 0.13 m

49. While aligning a hill road with a ruling gradient of 6 percent, a horizontal curve of radius 60 m is encountered. The compensated gradient at the curve is

- (A) 4.5% (B) 4.75%
(C) 5.0% (D) 5.25%

50. The maximum number of points of conflicts in two-way two-lane road meeting at right angle is

- (A) 8 (B) 12
(C) 24 (D) 32

51. One of the following methods is not used in conducting speed and delay studies. That method is

- (A) floating car method
(B) speed and delay method
(C) interview technique
(D) elevated observation

52. A vehicle of weight 2.0 tonne skids through a distance equal to 40 m before colliding with another parked vehicle of weight 1.0 tonne. After collision, both the vehicles skid through a distance equal to 12 m before stopping. If the average coefficient of friction is 0.5, both the vehicles after collision will move together with a speed of

- (A) 16.3 m/s
(B) 25.6 m/s
(C) 35.4 m/s
(D) 10.8 m/s

53. Fixed delay in highway is due to

- (A) road repairs
(B) pedestrian crossing the road
(C) traffic signal
(D) parked vehicle

54. The free mean speed on a roadway is found to be 80 kmph. Under stopped condition, the average spacing between vehicles is 6.9 m. The capacity flow is
- 3200 vehicles/hour/lane
 - 2900 vehicles/hour/lane
 - 2500 vehicles/hour/lane
 - 2200 vehicles/hour/lane
55. The type of stone aggregates not suitable for pavement is
- round aggregates
 - irregular aggregates
 - angular aggregates
 - flaky aggregates
56. The instantaneous speed of vehicles at a specified location is called
- spot speed
 - time mean speed
 - running speed
 - constant speed
57. What is the purpose of the first general principle of working from the whole to the part?
- It is easy to work from the whole to the part than to work from the part to the whole
 - To prevent accumulation of errors, if any
 - No compensating error of any kind is possible
 - All of the above
58. The distance between two points is measured by a 30 m long steel tape in suspension using a 15 kg pull. The cross-sectional area of the tape = 0.03 cm^2 and its total weight is 0.693 kg. The coefficient of thermal expansion, α for steel = 11×10^{-6} per $^{\circ}\text{C}$ and E for steel = $2.1 \times 10^6 \text{ kg/cm}^2$. The sag correction per tape length is
- + 0.00238 m
 - + 0.00267 m
 - 0.00238 m
 - 0.00267 m
59. The magnetic bearing of a line is $140^{\circ} 45'$. The true bearing of the line, if the magnetic declination is $5^{\circ} 30' \text{ W}$, is
- $146^{\circ} 15'$
 - $135^{\circ} 15'$
 - $39^{\circ} 15' \text{ E}$
 - $39^{\circ} 15' \text{ W}$
60. When a closed compass traverse is plotted, if the closing error is within the permissible limit, it is adjusted graphically by Bowditch's rule. The permissible limit of relative closing error is
- $\frac{1}{300}$
 - $\frac{1}{400}$
 - $\frac{1}{500}$
 - $\frac{1}{600}$
61. A 30-metre metric chain is divided into
- 100 links
 - 150 links
 - 200 links
 - 300 links

62. A man at a position 30 m above sea level just observes the peak of a hill. The peak of the hill is 200 m above sea level. Considering combined effect of curvature and refraction, the distance between the man and the peak of the hill is

- (A) 67.5 m (B) 72.3 m
(C) 75.6 m (D) 80.3 m

63. In a transit theodolite, the line of sight can be reversed by revolving the telescope through

- (A) 90° in horizontal plane
(B) 90° in vertical plane
(C) 180° in horizontal plane
(D) 180° in vertical plane

64. In ideal condition, in a closed traverse, the algebraic sum of latitudes and that of departures must be equal to

- (A) zero (B) 90°
(C) 180° (D) 270°

65. The ranges for WCB and QB are respectively

- (A) 0° to 90° and 0° to 180°
(B) 0° to 90° and 0° to 360°
(C) 0° to 180° and 0° to 360°
(D) 0° to 360° and 0° to 90°

66. Reverse curve should be avoided as far as possible for important tracks or highways because

- (A) it is difficult to construct a reverse curve
(B) setting out a reverse curve in the field is difficult
(C) superelevation cannot be provided at the point of reverse curvature
(D) All of the above

67. The length of a transition curve is given by

- (A) $\frac{V^2}{gR}$ (B) $\frac{V^3}{CR}$
(C) $\frac{\pi R\phi}{180^\circ}$ (D) $\frac{V^2}{CR}$

where

V = Speed in m/s

R = Radius of curve in m

C = Rate of change of radial acceleration in m/s³

g = Acceleration due to gravity

ϕ = Deflection angle

68. In the geometric design of highway, a type of vertical curve used to connect two gradients forming convex curve is known as

- (A) summit curve
(B) sag curve
(C) transition curve
(D) straight curve

69. If the plasticity index of a soil mass is zero, the soil is

- (A) clay (B) silt
(C) sand (D) clayey silt

70. Diffuse double layer in soil-water system is formed because surfaces of clay particles carry a net negative charge. This results from

- (A) isomorphous substitution and surface dissociation of hydroxyl ions
(B) absence of cations in the crystal lattice and adsorption of anions
(C) presence of organic matter
(D) any one combination of the factors mentioned in (A), (B) and (C) above

71. A soil with specific gravity of 2.65 has a void ratio of 0.65 in its loosest state and 0.30 in its densest state respectively. If the density index of the soil is 0.46, the dry density of the soil in its natural state is

- (A) 1.65 gm/cm^3
(B) 1.78 gm/cm^3
(C) 1.91 gm/cm^3
(D) 2.14 gm/cm^3

72. If a soil is in the semi-solid state, the liquidity index, I_L is

- (A) $I_L < 0$ (B) $0 < I_L < 1$
(C) $I_L = 1$ (D) $I_L > 1$

73. As per the IS plasticity chart, a soil is classified as CH soil. If the liquid limit of the soil is 55%, then the plastic limit of the soil is

- (A) $> 29.45\%$
(B) $< 29.45\%$
(C) $> 26.55\%$
(D) $< 26.55\%$

74. For the number of columns constructed in a row, the type of foundation provided is

- (A) raft (B) strap
(C) strip (D) footing

75. Vibratory rollers are most suited for

- (A) clayey soils
(B) silty clay soils
(C) granular soils
(D) All of the above

76. A laboratory compaction test on a soil having specific gravity equal to 2.65 and at a moisture content of 12%, corresponding to 5% air voids, will have a dry density of

- (A) 1.89 gm/cm^3
(B) 1.91 gm/cm^3
(C) 1.95 gm/cm^3
(D) 1.97 gm/cm^3

77. A sample of dense sand is tested in a triaxial shear with a confining pressure of 100 kN/m^2 . If the angle of internal friction of the sand is 38° , the deviator stress at failure is
- 220 kN/m^2
 - 320 kN/m^2
 - 420 kN/m^2
 - 520 kN/m^2
78. The unconfined compression test is a special type of
- vane shear test
 - unconsolidated-undrained triaxial test
 - unconsolidated-undrained direct shear test
 - drained triaxial shear test
79. As per IS:6403-1981, if the water table is just below the foundation level, in calculating the bearing capacity, the water table reduction factor, R_w will be
- $R_w = 0$
 - $R_w = 0.5$
 - $R_w = 1.0$
 - $0.5 < R_w < 1.0$
80. The method of boring adopted in boundary and gravelly strata is
- auger boring
 - wash boring
 - percussion boring
 - rotary boring
81. As per the IS codal guidelines for depth of investigation, in case rock formation is encountered prior to the termination depth, the boreholes may be extended, in rock strata, by minimum
- 3 m
 - 4 m
 - 5 m
 - 7 m
82. The depth interval between the bottom of the standard penetration test and next undisturbed sampling/any other test shall not be less than
- 200 mm
 - 300 mm
 - 400 mm
 - 500 mm
83. In soft cohesive soil and very loose cohesionless soil, especially below groundwater table, the undisturbed sample of soil collected is
- block sample
 - thin-walled sample
 - piston sample
 - core sample collected in core barrel
84. For undisturbed sampling in soft sensitive clays, the preferred area ratio of the sampler should be
- 10 percent or less
 - 12.5 percent or less
 - 15 percent or less
 - 20 percent or less

85. The angle of repose is the maximum angle which the outer face of the soil mass makes

- (A) with the horizontal
- (B) with the vertical
- (C) with the perpendicular to the inclined plane of soil
- (D) None of the above

86. A plate load test carried out on a loose sandy soil, under a pressure of 100 kN/m^2 , gives rise to a final equilibrium settlement of 5 mm. The plate size is $45 \text{ cm} \times 45 \text{ cm}$. If a footing of size $1 \text{ m} \times 1 \text{ m}$ is loaded with a pressure of 100 kN/m^2 , resting on the same soil, the final equilibrium settlement will be

- (A) 7.2 mm
- (B) 8.2 mm
- (C) 9.2 mm
- (D) None of the above

87. The intensity of vertical stress, induced at a point 5 m below a point load of 200 kN, acting on the surface of a soil mass with a large lateral restraint, so that it does not undergo any lateral strain, is

- (A) 3.8 kN/m^2 (B) 2.5 kN/m^2
- (C) 0.5 kN/m^2 (D) 1.5 kN/m^2

88. The ultimate load capacity of a uniform shaft bored and cast-in-situ pile, 450 mm in diameter and 6.0 m long, in a cohesive soil deposit with an undrained shear strength of 30 kN/m^2 , is

- (A) 254.5 kN (B) 297.4 kN
- (C) 42.9 kN (D) 118.9 kN

89. When a pile in a group, designed for a certain safe load, is found to fall just short of the load required to be carried by it, the maximum overload, allowed above the allowable load on a single pile of the pile group, is

- (A) up to 10 percent
- (B) up to 15 percent
- (C) up to 20 percent
- (D) up to 25 percent

90. The sub-soil for a proposed 2-storeyed residential building is found to be 6 m thick black cotton soil, underlain by stiff sandy clay. The most suitable foundation for the proposed structure may be

- (A) shallow isolated or combined footing
- (B) shallow strip footing
- (C) raft foundation
- (D) under-reamed pile

91. The measurement of non-biodegradable organic content is usually carried out in terms of

- (A) BOD (B) COD
- (C) TOC (D) DO

92. Pick out the incorrect answer :

Automobiles, without a catalytic converter, chiefly emit

- (A) carbon monoxide
- (B) carbon dioxide
- (C) hydrocarbons
- (D) oxides of nitrogen

93. A rainfall is considered as acidic if its pH value is less or equal to
 (A) 5 (B) 5.6
 (C) 6.5 (D) 7
94. If nitrate has crossed the limit of acceptance, which disease may cause?
 (A) Polio
 (B) Bone cancer
 (C) Lung cancer
 (D) Blue baby
95. The amount of money which can be obtained at any particular time from the open market, if the property is put for sale, is called
 (A) salvage value
 (B) market value
 (C) book value
 (D) rateable value
96. After the useful life of a building, the scrap value is considered as
 (A) 5% of cost of the building
 (B) 10% of cost of the building
 (C) 15% of cost of the building
 (D) 20% of cost of the building
97. PERT technique of network analysis is mainly useful for
 (A) deterministic activities
 (B) large and complex projects
 (C) research and development projects
 (D) small projects
98. In the network method of construction management, slack or float is defined as
 (A) difference between latest starting time and earliest starting time for an event
 (B) difference between latest finishing time and earliest finishing time for an event
 (C) difference between earliest finishing time and earliest starting time for an event
 (D) difference between latest finishing time and latest starting time for an event
99. In project crashing, rent and overheads are treated as
 (A) direct costs
 (B) indirect costs
 (C) significant costs
 (D) insignificant costs
100. Pick out the incorrect statement.
 (A) Events with zero slacks are critical.
 (B) If there is negative slack found in any activity, it may cause delay in the completion of the project.
 (C) Critical path is the line connecting the activities with zero slack.
 (D) Critical path is normally shown with thick dotted lines in the network.

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